

Chapter 1

First Order Ordinary Differential Equations

Learning Objectives

By the end of this chapter, students should be able to

1. define an Ordinary Differential Equation (ODE);
2. determine the order of an ODE;
3. distinguish between general and particular solutions of an ODE;
4. distinguish between separable and linear ODEs;
5. solve first order ODEs using the variable separation method or the integrating factor method;
6. solve engineering problems modelled by first order ODEs.

Differential Equations (DE) are indispensable in the study of engineering and modern science. Many engineering problems are modelled into one of two types of differential equations, namely, Ordinary Differential Equations (ODEs) and Partial Differential Equations (PDEs).

We begin this chapter by introducing the ODEs and their orders in Section 1.1, followed by discussing two methods in solving first order ODEs: the variable separation method in Section 1.2, and the integrating factor method in Section 1.3. In Section 1.4, we look at some engineering problems modelled by first order ODEs. Finally, we close the chapter with exercises in Section 1.5, where student gets to solve first order ODEs using the two methods discussed, as well as a taste of more advanced techniques.

1.1 Introduction

Definition 1.1.1 (Ordinary Differential Equations)

An **Ordinary Differential Equation** (ODE) is an equation involving the derivatives of an output variable y with respect to the input variable x .

The **order** of an ODE is the order of the highest derivative in the equation.

The following are examples of ODEs with their orders.

Example 1.1.2

- (a) $\frac{dy}{dx} = 3x^2$ First order
- (b) $xy' + y = 2x$ First order
- (c) $y'' = y'$ Second order
- (d) $\frac{dx}{dt} = tx^2$ First order

Definition 1.1.3 (Solution of a DE)

A **solution** of a DE is an equation that satisfies the DE but does not involve any derivatives.

There are two possible types of solutions in DEs:

1. The **general solution** of an n -th order DE is a solution that involves n arbitrary constants. It includes all possible solutions of the DE.
2. A **particular solution** of a DE is a solution that does NOT involve any arbitrary constants. This happens when initial conditions are known.

A particular solution of an ODE is obtained when additional information are given, such as initial value(s). ODEs where initial conditions are known are called **Initial Value Problems (IVP)**.

Example 1.1.4

	ODE/IVP	Solution	Type
(a)	$\frac{dy}{dx} = 3x^2$	$y = x^3 + C$	General solution
(b)	$y'' = y$	$y = C_1 e^x + C_2 e^{-x}$	General solution
(c)	$\begin{cases} xy' + y = 2x \\ y(1) = 0 \end{cases}$	$y = x - \frac{1}{x}$	Particular solution
(d)	$\begin{cases} y'' = 2y' \\ y(0) = 0 \\ y'(0) = -1 \end{cases}$	$y = 1 - e^{2x}$	Particular solution

From Example 1.1.4(a) and (b), we observe that the general solution of a first order ODE contains one arbitrary constant, while that of a second order ODE contains two arbitrary constant.

From Example 1.1.4(c) and (d), we observe that a first order IVP includes one initial value, while a second order IVP includes two initial values.

The next few examples show how general solutions to ODEs can be found by noting that $\frac{dy}{dx} = f(x)$ implies $y = \int f(x) dx$.

Example 1.1.5

Solve $y' = \sin 2x$.

Solution

$$y' = \sin 2x \Rightarrow y = \int \sin 2x \, dx \Rightarrow y = -\frac{1}{2} \cos 2x + C.$$

Example 1.1.6

Solve $x \frac{dy}{dx} - 1 = 2x^2$.

Solution

$$x \frac{dy}{dx} - 1 = 2x^2 \Rightarrow x \frac{dy}{dx} = 2x^2 + 1 \Rightarrow \frac{dy}{dx} = 2x + \frac{1}{x}$$

$$\Rightarrow y = \int \left(2x + \frac{1}{x} \right) dx$$

$$\Rightarrow y = x^2 + \ln|x| + C.$$

Example 1.1.7

Solve the IVP: $x \frac{dy}{dx} - 1 = 2x^2$, where $y(1) = 0$.

Solution

From the previous example, the general solution of $x \frac{dy}{dx} - 1 = 2x^2$ is

$$y = x^2 + \ln|x| + C.$$

$$y(1) = 0 \Rightarrow 0 = 1^2 + \ln|1| + C \Rightarrow C = -1$$

The particular solution is $y = x^2 + \ln|x| - 1$.

1.2 Separable ODEs

Definition 1.2.1 (Separable ODE)

A **separable ODE** is an ODE that can be written as $\frac{dy}{dx} = f(x) g(y)$.

Example 1.2.2

Identify which of the following ODEs are separable:

(a) $\frac{dy}{dx} = xy$

(b) $y' = x^2 e^{-y}$

(c) $\frac{dy}{dx} + 2y = 2x$

Solution

- (a) Separable.
- (b) Separable.
- (c) Non-separable.

Procedure in Solving Separable ODE

Step 1. Group $g(y)$ with dy and $f(x)$ with dx to give $\frac{dy}{g(y)} = f(x) dx$.

Step 2. Integrate both sides: $\int \frac{1}{g(y)} dy = \int f(x) dx$ to give the general solution.

Example 1.2.3

Solve $\frac{dy}{dx} = xy$.

Solution

Step 1. $\frac{dy}{dx} = xy \Rightarrow \frac{1}{y} dy = x dx$

Step 2. $\int \frac{1}{y} dy = \int x dx \Rightarrow \ln |y| = \frac{1}{2}x^2 + C.$

Example 1.2.4

Solve $y' = y$.

Solution

Step 1. $\frac{dy}{dx} = y \Rightarrow \frac{1}{y} dy = dx$

Step 2. $\int \frac{1}{y} dy = \int dx \Rightarrow \ln |y| = x + C$ or $y = C_1 e^x.$

Example 1.2.5

Solve $\frac{dy}{dx} = e^{2x} \sec y$.

Solution

Step 1. $\frac{dy}{dx} = e^{2x} \sec y \Rightarrow \frac{1}{\sec y} dy = e^{2x} dx$

Step 2. $\int \cos y dy = \int e^{2x} dx \Rightarrow \sin y = \frac{1}{2}e^{2x} + C. \left(\text{note: } \sec y = \frac{1}{\cos y}\right)$

Example 1.2.6

Solve $\frac{dy}{dx} = x^2 e^{-y}$.

Solution

Step 1. $\frac{dy}{dx} = x^2 e^{-y} \Rightarrow e^y dy = x^2 dx$

Step 2. $\int e^y dy = \int x^2 dx \Rightarrow e^y = \frac{1}{3}x^3 + C.$

Example 1.2.7

Solve the IVP: $y' = \frac{x^2}{e^y}$, where $y(0) = 1$.

Solution

From the previous example, the general solution of $y' = \frac{x^2}{e^y}$ is

$$e^y = \frac{1}{3}x^3 + C.$$

$$y(0) = 1 \Rightarrow e^1 = 0 + C \Rightarrow C = e$$

The particular solution is $e^y = \frac{1}{3}x^3 + e$.

1.3 Linear ODE

A first order linear ODE is an ODE of the form $P_1 \frac{dy}{dx} + P_0 y = F$, where P_0 , P_1 , y , and F are functions of x , with $P_1 \neq 0$. The solution can be found by using the **integrating factor**. As a start, we rewrite the ODE in standard form.

Definition 1.3.1 (Standard Form of First Order Linear ODE)

The **standard form of a first order linear ODE** is $\frac{dy}{dx} + P(x)y = Q(x)$.

Remark. In Definition 1.3.1, P and Q are functions of x . If either P or Q is zero, then the ODE is a separable ODE.

Procedure in Solving First Order Linear ODE

Step 1. Write the ODE in standard form and identify $P(x)$ and $Q(x)$.

Step 2. Evaluate and simplify the integrating factor $I(x) = e^{\int P(x) dx}$.

Step 3. The general solution is $I(x)y = \int I(x)Q(x) dx$.

Example 1.3.2

Solve $\frac{dy}{dx} + 2y = e^{-x}$.

Solution

Step 1. $P(x) = 2$ and $Q(x) = e^{-x}$

Step 2. $I(x) = e^{\int 2 dx} = e^{2x}$

Step 3. The general solution is $e^{2x}y = \int e^{2x}e^{-x}dx = \int e^x dx$

$$\Rightarrow e^{2x}y = e^x + C$$

We may use the laws of exponents and radicals to simplify the integrating factor. In the following examples, we note that $e^{\ln A} = A$.

Example 1.3.3

Solve $\frac{dy}{dx} + \frac{2y}{x} = 1$, where $x > 0$.

Solution

Step 1. $P(x) = \frac{2}{x}$ and $Q(x) = 1$

Step 2. $I(x) = e^{\int \frac{2}{x} dx} = e^{2 \ln x} = e^{\ln x^2} = x^2$

Step 3. The general solution is $x^2 y = \int x^2 dx$

$$\Rightarrow x^2 y = \frac{1}{3} x^3 + C$$

Example 1.3.4

Solve $\frac{dy}{dx} + \left(1 + \frac{1}{x}\right)y = \frac{1}{x}$, where $x > 0$.

Solution

Step 1. $P(x) = 1 + \frac{1}{x}$ and $Q(x) = \frac{1}{x}$

Step 2. $I(x) = e^{\int \left(1 + \frac{1}{x}\right) dx} = e^{x + \ln x} = e^x e^{\ln x} = x e^x$

Step 3. The general solution is $x e^x y = \int (x e^x) \left(\frac{1}{x}\right) dx = \int e^x dx$

$$\Rightarrow x e^x y = e^x + C$$

1.4 Applications

In this section, we discuss engineering problems modelled by first order DEs. The modelling processes are briefly described for examples in the life sciences, mechanics, and electronics.

Applications in the Life Sciences

The exponential growth/decay model is often used in describing the growth/decay of a substance. The model is based on the assumption that the derivative Q' is directly proportional to the quantity Q . In other words,

$$\frac{dQ}{dt} = kQ,$$

where k is a non-zero constant. If $k > 0$, then it is a growth model; otherwise it is a decay model.

The growth/decay model above is a separable ODE. To solve for the particular solution, we consider the initial quantity $Q(0)$.

Example 1.4.1

In an experiment of testing a new drug against a certain type of bacteria, a scientist modelled the decay of the bacteria population P by the differential equation

$$\frac{dP}{dt} = -0.2457P$$

Suppose the initial population of the bacteria in a culture is approximately 2 (in million units). Express the bacteria population P in terms of time t (in days).

Solution

$$\frac{dP}{dt} = -0.2457P \Rightarrow \frac{1}{P} dP = -0.2457 dt$$

$$\Rightarrow \int \frac{1}{P} dP = -0.2457 \int dt$$

$$\Rightarrow \ln P + C = -0.2457 t$$

$$P(0) = 2 \Rightarrow \ln 2 + C = 0 \Rightarrow C = -\ln 2$$

$$\text{Thus, } \ln P - \ln 2 = -0.2457 t \quad \text{or} \quad P = 2 e^{-0.2457 t}.$$

Applications in Mechanics

In classical mechanics, we study the motion of an object subjected to various forces. The **Newton's Law** $F = ma$ serves as the principle in modelling mechanical problems. Here, F denotes the total forces involved, and m denotes the mass of the object. Moreover, a denotes acceleration of the object, defined as the derivative of the velocity, v , with respect to time, t :

$$a = \frac{dv}{dt}.$$

Alternatively, if the velocity v is regarded as a function of x , the displacement of the object, then the acceleration can be written as follows:

$$a = v \frac{dv}{dx}.$$

We shall discuss two examples of mechanics problems modelled by differential equations involving $\frac{dv}{dt}$ or $\frac{dv}{dx}$, and solve for the particular solutions by considering the initial velocity $v(0)$.

Example 1.4.2

A ball of unit mass moves horizontally in a straight line with initial velocity 1 m/s. The motion of the ball satisfies the differential equation

$$-\frac{1}{2}v^3 = \frac{dv}{dt}.$$

Express v in terms of t .

Solution

$$-\frac{1}{2}v^3 = \frac{dv}{dt} \Rightarrow dt = \frac{-2}{v^3} dv$$

$$\Rightarrow \int dt = -2 \int v^{-3} dv$$

$$\Rightarrow t = v^{-2} + C$$

At $t = 0$, the velocity is $v = 1$ m/s. Hence, $0 = 1^{-2} + C \Rightarrow C = -1$.

And thus, $t = v^{-2} - 1$ or $v = \sqrt{\frac{1}{t+1}}$.

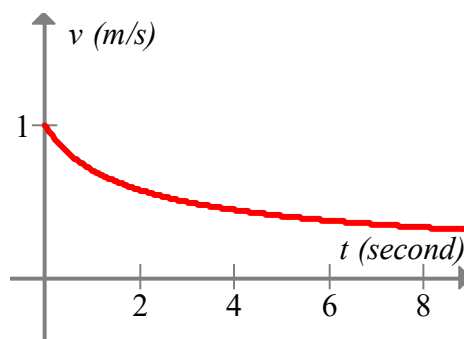


Figure 1.1. Graph of v in terms of t in Example 1.4.2

Example 1.4.3

A particle of unit mass is projected vertically upwards with initial velocity 1 m/s, and subjected to earth's gravity. The motion of the particle satisfies the differential equation

$$v \frac{dv}{dx} = -g - \frac{1}{3} g v^2,$$

where x denotes the vertical (upward) displacement of the particle, and $g \approx 9.8 \text{ m/s}^2$ is the constant of gravitational acceleration.

- (a) Express x in terms of v .
 (b) Calculate the maximum displacement of the particle.

Solution

$$(a) \quad v \frac{dv}{dx} = -g - \frac{1}{3} g v^2 \Rightarrow v \frac{dv}{dx} = -g \left(1 + \frac{v^2}{3} \right)$$

$$\Rightarrow \frac{v}{-g \left(1 + \frac{v^2}{3} \right)} dv = dx$$

$$\Rightarrow \frac{1}{-g} \int \frac{3v}{3 + v^2} dv = \int dx$$

$$\Rightarrow -\frac{3}{2g} \ln(3 + v^2) + C = x$$

At $x = 0$, the velocity is $v = 1$. So, $-\frac{3}{2g} \ln(3 + 1^2) + C = 0 \Rightarrow C = \frac{3}{2g} \ln 4$.

Thus, $-\frac{3}{2g} \ln(3 + v^2) + \frac{3}{2} \ln 4 = x \Rightarrow x = \frac{3}{2g} \ln \left(\frac{4}{3+v^2} \right)$.

- (b) The maximum displacement is obtained when $v = 0$.

Thus, $x_{\max} = \frac{3}{2g} \ln \left(\frac{4}{3} \right) \approx 0.0440 \text{ m}$.

Applications in Electronics

In circuit analysis, we study how interactions of electronics components affect the behaviour of the current and voltage in an electrical circuit. The **Kirchoff's Voltage Law (KVL)** serves as the principle in modelling electronics problems. KVL states that the sum of the voltage across each component in a series circuit is equal to the voltage source, i.e.

$$V_R + V_L + V_C = V_{\text{source}}.$$

where V_R , V_L , and V_C denotes the voltage across resistor, inductor, and capacitor, respectively, and they satisfy the following properties:

$$V_R = iR, \quad V_L = L \frac{di}{dt}, \quad V_C = \frac{q}{C}.$$

Here, i denotes the current (in amperes), R the resistance (in ohms), L the inductance (in henrys), q the charge (in coulombs), and C the capacitance (in farads), and t the time (in seconds).

In a closed series circuit, the same current passes through each electrical component in the circuit. Moreover, the current is defined as the derivative of the charge with respect to time:

$$i = \frac{dq}{dt}.$$

We shall discuss three examples of electronics problems modelled by linear differential equations involving $\frac{di}{dt}$ or $\frac{dq}{dt}$, and solve for their particular solutions by considering initial current $i(0)$ or initial charge $q(0)$.

Example 1.4.4

A series circuit consists of an inductor of inductance 1 H, a resistor of resistance 1 k Ω and a 9V battery. The circuit is closed at time $t = 0$ and satisfies the differential equation

$$\frac{di}{dt} + 1000i = 9.$$

Express the current i in terms of time t .

Solution

$\frac{di}{dt} + 1000i = 9$ is a standard 1st order linear ODE

$$P(t) = 1000 \text{ and } Q(t) = 9$$

$$I(t) = e^{\int 1000 dt} = e^{1000t}$$

$$e^{1000t}i = \int e^{1000t}(9)dt = \frac{9}{1000}e^{1000t} + C$$

$$\Rightarrow i = 0.009 + Ce^{-1000t}$$

Since the circuit is closed at time $t = 0$, $i(0) = 0$, and hence,

$$0 = 0.009 + Ce^0 \Rightarrow C = -0.009$$

Thus, $i = 0.009 - 0.009e^{-1000t}$ or $i = 0.009(1 - e^{-1000t})$.

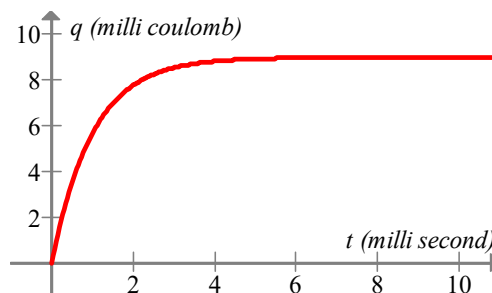


Figure 1.2. Graph of q in terms of t in Example 1.4.4

Example 1.4.5

A series circuit consists of a capacitor of capacitance 0.01 F , a resistor of resistance $0.1 \text{ k}\Omega$ and a voltage source $V_{\text{source}} = 200 e^{-t} \sin t$. The capacitor is charged with an initial charge $q(0) = 1 \text{ C}$ and satisfies the differential equation

$$\frac{dq}{dt} + q = 2 e^{-t} \sin t.$$

Express the charge q in terms of time t .

Solution

$$P(t) = 1 \text{ and } Q(t) = 2 e^{-t} \sin t$$

$$I(t) = e^{\int 1 dt} = e^t$$

$$e^t q = \int e^t (2 e^{-t} \sin t) dt$$

$$\Rightarrow e^t q = 2 \int \sin t dt = -2 \cos t + C$$

$$\Rightarrow q = -2e^{-t} \cos t + Ce^{-t}$$

$$q(0) = 1 \Rightarrow 1 = -2e^0 \cos 0 + Ce^0 \Rightarrow C = 3$$

Thus, $q = -2e^{-t} \cos t + 3e^{-t}$ or $q = e^{-t}(3 - 2 \cos t)$.

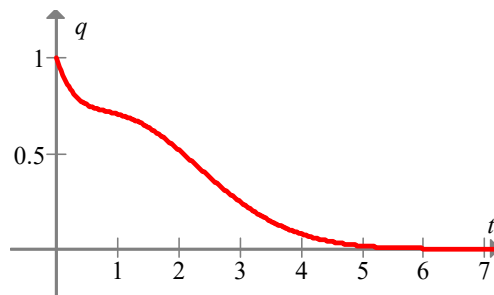


Figure 1.3. Graph of q in terms of t in Example 1.4.5

Example 1.4.6

A capacitor of capacitance 10 mF is fully charged with initial voltage $V_C(0) = 10$ V. It is then discharged through a resistor of resistance 0.1 k Ω and satisfies

$$\frac{dV_C}{dt} + V_C = 0.$$

Express the voltage across the capacitor V_C in terms of time t .

Solution

$$\frac{dV_C}{dt} + V_C = 0 \Rightarrow \frac{dV_C}{dt} = -V_C$$

$$\Rightarrow \frac{1}{V_C} dV_C = -dt$$

$$\Rightarrow \int \frac{1}{V_C} dV_C = - \int dt$$

$$\Rightarrow \ln V_C + k = -t$$

$$V_C(0) = 10 \Rightarrow \ln 10 + k = 0 \Rightarrow k = -\ln 10$$

$$\text{Thus, } \ln V_C - \ln 10 = -t \Rightarrow \ln(0.1 V_C) = -t \text{ or } V_C = 10e^{-t}.$$

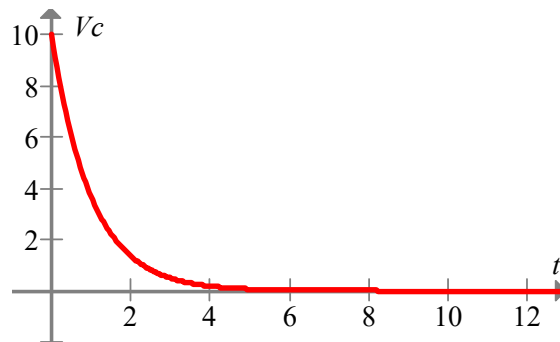


Figure 1.4. Graph of V_C in terms of t in Example 1.4.6

1.5 Exercises

Part A: Baseline Problems

1. Solve the following first order separable ODEs (see Section 1.2).

(a) $\frac{dy}{dx} = \cos x$

(f) $(x^2 + 1)\frac{dy}{dx} = x \sec y$

(b) $\frac{dy}{dx} = \frac{e^x}{3y^2}$

(g) $x\frac{dy}{dx} = \sqrt{x} + x \sin x$

(c) $x\frac{dy}{dx} = 2y$

(h) $\frac{dx}{dt} = e^{2t}$

(d) $\cos^2 x \frac{dy}{dx} - 1 = 0$

(i) $\frac{dx}{dt} = \sqrt{xt}$

(e) $x^2 + y^2 \frac{dy}{dx} = 1$

(j) $\frac{dx}{dt} = \frac{e^{2t-x}}{x}$

2. Solve the following first order linear ODEs (see Section 1.3).

(a) $\frac{dy}{dx} + y = -e^{-x}$

(f) $x\frac{dy}{dx} + (1+x)y = e^{-x}$, where $x > 0$

(b) $\frac{dy}{dx} + \frac{2}{x}y = -x$

(g) $\sin x \frac{dy}{dx} + y \cos x = \sin 2x$, where $0 < x < \pi$

(c) $\frac{dy}{dx} + y = e^{-x} \sin 2x$

(h) $(x+1)\frac{dy}{dx} + y = x+2$, where $x > -1$

(d) $\frac{dy}{dx} - 2y = e^{2x} \cos 2x$

(i) $(1-x^2)\frac{dy}{dx} + 4y = \frac{3}{1+x}$

(e) $x\frac{dy}{dx} + 2y = \frac{\cos x}{x}$

(j) $(1+x^2)\frac{dy}{dx} + 2xy = \cos 4x$

Part B: Intermediate Problems

3. Solve the following Initial Value Problems (IVP) (see Example 1.2.7).

$$(a) \quad \begin{cases} \frac{dy}{dx} = \frac{1}{1+x^2} \\ y(0) = 1 \end{cases}$$

$$(b) \quad \begin{cases} \frac{dy}{dx} = \sec^2 x \\ y(0) = -1 \end{cases}$$

$$(c) \quad \begin{cases} y' = 4xy \\ y(0) = 2 \end{cases}$$

4. Solve the following Initial Value Problems (IVP).

$$(a) \quad \begin{cases} \frac{dy}{dx} - y = e^x \\ y(0) = 4 \end{cases}$$

$$(b) \quad \begin{cases} \frac{dy}{dx} + \frac{y}{x} = e^x, \text{ where } x > 0 \\ y(1) = 0 \end{cases}$$

$$(c) \quad \begin{cases} xy' + (1+x)y = e^{-x}, \text{ where } x > 0 \\ y(1) = 1 \end{cases}$$

Part C: Application Problems

(See Section 1.4)

5. In a chemical experiment, a tank is initially filled with 30 litres of pure water. A chemical substance is being poured into the tank at the rate of 5 litres/minute. The mixture is kept uniform by constant stirring, and flows out of the tank at the same rate as the chemical substance being poured in. The amount of the substance in the tank satisfies the differential equation

$$\frac{dx}{dt} = 10 - \frac{5x}{30},$$

where x denotes the amount of the substance at time t minutes.

- (a) Express x in term of t .
(b) How much substance is present in the tank after an hour of mixing?
6. A forensic officer examined a corpse found in an air-conditioned room of constant temperature 25°C . At the time of investigation, the corpse had a temperature of 35°C . According to Newton's Law of Cooling, the cooling body satisfies the differential equation

$$\frac{dx}{dt} = -0.1054(x - 25),$$

where x denotes the temperature of the cooling body (in $^{\circ}\text{C}$), and t the time (in hours).

- (a) Express x in term of t .
(b) Suppose the person died in the room and had a body temperature of 36°C at the time of death. How long had the body been in the room since it was found?

7. A white snooker ball of mass 0.16 kg travels at initial velocity 2 m/s in a horizontal direction towards a blue snooker ball located 0.5 m away. The motion of the white ball is subjected to resistance, and its velocity, v , satisfies the differential equation

$$\frac{dv}{dx} = -\frac{1}{0.32},$$

where x denotes the displacement of the white ball from its initial position.

- (a) Express v in terms of x .
- (b) Will the white ball hit the blue ball?

8. A ball of unit mass is projected vertically upwards with an initial velocity 3 m/s. The motion of the ball is subjected to earth's gravity and air resistance, and its velocity, v , satisfies the differential equation

$$\frac{dv}{dt} = -g(1 + v),$$

where $g \approx 9.8 \text{ m/s}^2$ is the constant of gravitational acceleration.

- (a) Express v in terms of t .
- (b) How long does it take the ball to reach maximum height?

9. A 5-volt voltage source is connected to a 1 H inductor and a 0.5 k Ω resistor in series. The current i on the circuit satisfies the differential equation

$$\frac{di}{dt} + 500i = 5.$$

Given zero initial current in the circuit, express the current i in terms of t .

10. A fully charged 0.05 F capacitor is discharged through a 1 k Ω resistor. The voltage across the capacitor V_C satisfies the differential equation

$$50 \frac{dV_C}{dt} + V_C = 0.$$

Given an initial voltage of 25 volts across the capacitor, express the V_C in terms of t .

11. A series circuit consists of a capacitor of capacitance 0.0625 F, a resistor of resistance 0.8 k Ω and a voltage source of 24 volts. The circuit is closed and satisfies the differential equation

$$10 \frac{dq}{dt} + 2q = 3.$$

Given that the capacitor has an initial charge of 0.5 C, express the charge q in terms of t .